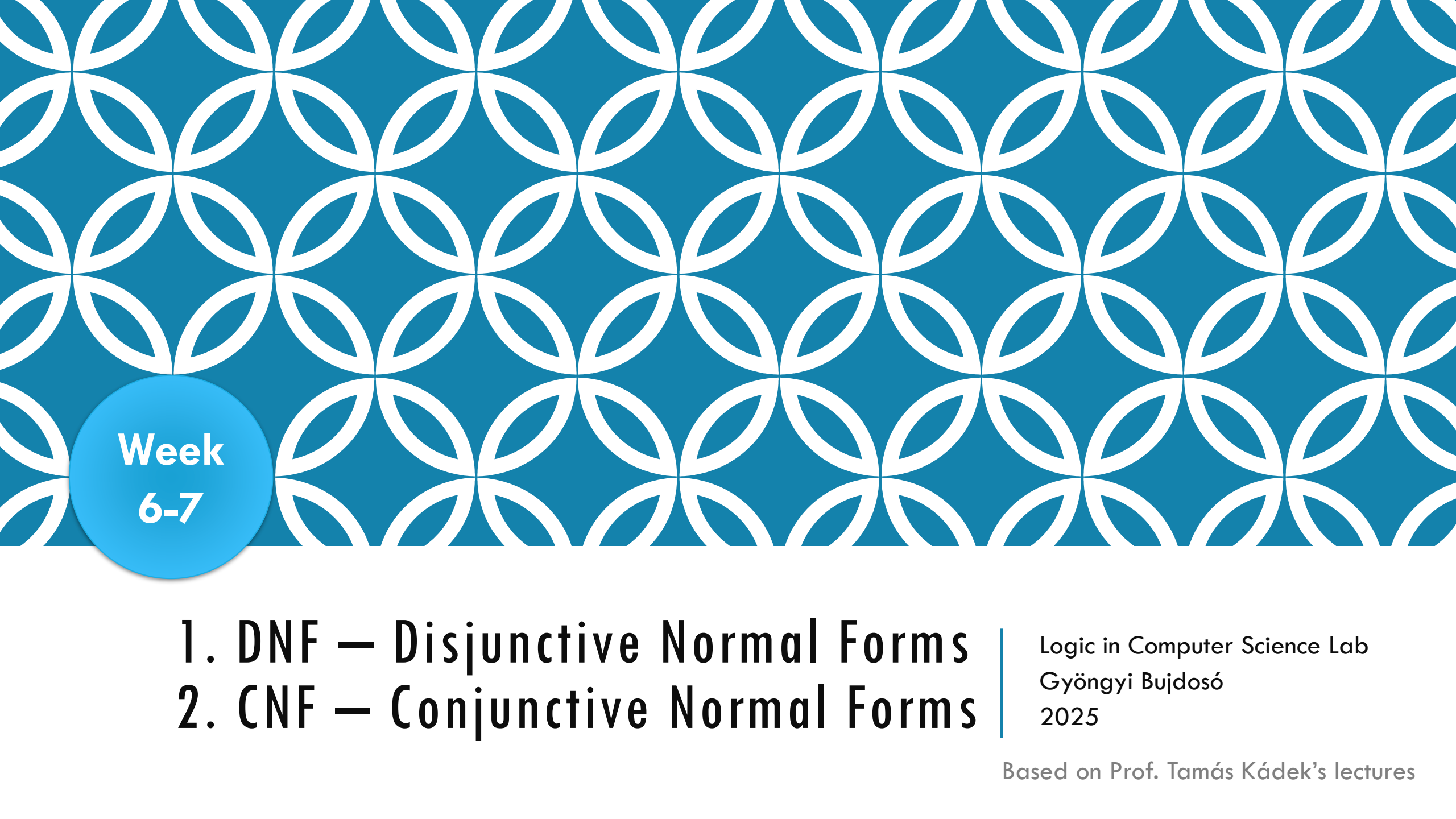




**Week
6-7**

- 1. DNF — Disjunctive Normal Forms**
- 2. CNF — Conjunctive Normal Forms**

Logic in Computer Science Lab
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Based on Prof. Tamás Kádek's lectures

Notations and definitions

Literal: an atomic formula or a negated atomic formula

Elementary disjunction: Disjunction of literals

Elementary conjunction: Conjunction of literals

Normal form: every formula can be expressed in these forms:

$$(a \wedge b \wedge \neg c) \vee (\neg a \wedge b \wedge c) \vee (a \wedge \neg b \wedge \neg c) \quad - \text{DNF}$$

$$(a \vee b \vee \neg c) \wedge (\neg a \vee b \vee c) \wedge (a \vee \neg b \vee \neg c) \quad - \text{CNF}$$

Disjunctive Normal Form (DNF)

DNF – Disjunctive Normal Form:

A *disjunctive normal form* (DNF):

- is an elementary conjunction, or
- a disjunction of an elementary conjunction and a disjunctive normal form.

$e_1 \vee e_2 \vee \cdots \vee e_n$ where $n \geq 1$ and e_i is an elementary conjunction for all $i = 1, \dots, n$

Samples for DNF

DNF

A

$A \vee B$

$A \vee \neg B$

$A \wedge \neg B$

$(A \wedge \neg B) \vee (\neg A \wedge \neg B)$

Not in
DNF

$\neg(A \wedge B) \vee C,$

$(A \vee B) \wedge C,$

$(A \vee B) \supset C,$

But in these forms,
they are in DNF

$\neg A \vee \neg B \vee C$

$(A \wedge C) \vee (B \wedge C)$

$(\neg A \wedge \neg B) \vee (C)$

Conjunctive Normal Form (CNF)

CNF – Conjunctive Normal Form:

A *conjunctive normal form* (CNF):

- is an elementary disjunction, or
- a conjunction of an elementary disjunction and a conjunctive normal form.

$e_1 \wedge e_2 \wedge \cdots \wedge e_n$ where $n \geq 1$ and e_i is an elementary disjunction for all $i = 1, \dots, n$

Samples for CNF

CNF

$$A \vee B$$

$$A \vee \neg B$$

$$A \wedge \neg B$$

$$(A \vee \neg B) \wedge (\neg A \vee \neg B)$$

Not in
CNF

$$\neg(A \vee B) \wedge C,$$

$$(A \wedge B) \vee C,$$

$$(A \vee B) \supset C$$

But in these forms,
they are in CNF,
and logically equivalent to
the previous ones

$$\neg A \wedge \neg B \wedge C$$

$$(A \vee C) \wedge (B \vee C)$$

$$(\neg A \vee C) \wedge (\neg B \vee C)$$

Properties of conjunction

- Commutative: $(A \wedge B) \Leftrightarrow (B \wedge A)$
for all $A, B \in \text{Form}$.
- Associative: $(A \wedge (B \wedge C)) \Leftrightarrow ((A \wedge B) \wedge C)$ for all $A, B, C \in \text{Form}$.
- Idempotent: $(A \wedge A) \Leftrightarrow A$ for all $A \in \text{Form}$.
- $(A \wedge B) \models A, (A \wedge B) \models B$
- The law of contradiction: $\models \neg(A \wedge \neg A)$
- If $A, A_1, A_2, \dots, A_n \in \text{Form}$, then
 - ▶ the set $\{A_1, A_2, \dots, A_n\}$ is satisfiable, iff the formula $A_1 \wedge A_2 \wedge \dots \wedge A_n$ is satisfiable
 - ▶ the set $\{A_1, A_2, \dots, A_n\}$ is unsatisfiable, iff the formula $A_1 \wedge A_2 \wedge \dots \wedge A_n$ is unsatisfiable
 - ▶ $\{A_1, A_2, \dots, A_n\} \models A$ iff $A_1 \wedge A_2 \wedge \dots \wedge A_n \models A$,
 - ▶ $\{A_1, A_2, \dots, A_n\} \models A$ iff $A_1 \wedge A_2 \wedge \dots \wedge A_n \wedge \neg A$ is unsatisfiable.

Properties of disjunction:

- Commutative: $(A \vee B) \Leftrightarrow (B \vee A)$ for all $A, B \in \text{Form}$.
- Associative: $(A \vee (B \vee C)) \Leftrightarrow ((A \vee B) \vee C)$ for all $A, B, C \in \text{Form}$.
- Idempotent: $(A \vee A) \Leftrightarrow A$ for all $A \in \text{Form}$.
- $A \models (A \vee B)$ for all $A, B \in \text{Form}$.
- $\{(A \vee B), \neg A\} \models B$
- The law of excluded middle: $\models (A \vee \neg A)$
- Conjunction and disjunction are dual truth functors.
 - ▶ replacing disjunction-conjunction, true-false
- Two laws of distributivity:
 - ▶ $(A \vee (B \wedge C)) \Leftrightarrow ((A \vee B) \wedge (A \vee C))$
 - ▶ $(A \wedge (B \vee C)) \Leftrightarrow ((A \wedge B) \vee (A \wedge C))$
- Properties of absorption
 - ▶ $(A \wedge (B \vee A)) \Leftrightarrow A$
 - ▶ $(A \vee (B \wedge A)) \Leftrightarrow A$

De Morgan's laws

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$$

$$\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$



Augustus De Morgan (1806–1871)

Expressibility

$$(A \supset B) \Leftrightarrow \neg(A \wedge \neg B)$$

$$(A \supset B) \Leftrightarrow (\neg A \vee B)$$

$$(A \wedge B) \Leftrightarrow \neg(A \supset \neg B)$$

$$(A \vee B) \Leftrightarrow (\neg A \supset B)$$

$$(A \vee B) \Leftrightarrow \neg(\neg A \wedge \neg B)$$

$$(A \wedge B) \Leftrightarrow \neg(\neg A \vee \neg B)$$

$$(A \equiv B) \Leftrightarrow ((A \supset B) \wedge (B \supset A))$$

De Morgan Laws

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$$

$$\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$

Distributivity:

- $A \vee (B \wedge C) \Leftrightarrow (A \vee B) \wedge (A \vee C)$
- $A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$

$$(A \wedge B) \supset C \Leftrightarrow A \supset (B \supset C)$$

$$A \supset B \Leftrightarrow \neg B \supset \neg A$$

$$A \supset (B \supset C) \Leftrightarrow (A \supset B) \supset (A \supset C)$$

$$(A \vee B) \supset C \Leftrightarrow (A \supset C) \wedge (B \supset C)$$

Expressibility

- $A \supset B \Leftrightarrow \neg(A \wedge \neg B)$
- $A \supset B \Leftrightarrow (\neg A \vee B)$
- $A \wedge B \Leftrightarrow \neg(A \supset \neg B)$
- $A \vee B \Leftrightarrow (\neg A \supset B)$
- $A \vee B \Leftrightarrow \neg(\neg A \wedge \neg B)$
- $A \wedge B \Leftrightarrow \neg(\neg A \vee \neg B)$
- $A \equiv B \Leftrightarrow (A \supset B) \wedge (B \supset A)$

Rules for expressibility (for equivalent transformations)

$$\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$$

$$(A \supset B) \Leftrightarrow \neg(A \wedge \neg B)$$

$$(A \supset B) \Leftrightarrow (\neg A \vee B)$$

$$(A \wedge B) \Leftrightarrow \neg(A \supset \neg B)$$

$$(A \vee B) \Leftrightarrow (\neg A \supset B)$$

$$(A \vee B) \Leftrightarrow \neg(\neg A \wedge \neg B)$$

$$(A \wedge B) \Leftrightarrow \neg(\neg A \vee \neg B)$$

$$(A \equiv B) \Leftrightarrow ((A \supset B) \wedge (B \supset A))$$

$$(A \vee (B \wedge C)) \Leftrightarrow ((A \vee B) \wedge (A \vee C))$$

$$(A \wedge (B \vee C)) \Leftrightarrow ((A \wedge B) \vee (A \wedge C))$$

$$(A \wedge (B \vee A)) \Leftrightarrow A$$

$$(A \vee (B \wedge A)) \Leftrightarrow A$$

De Morgan Laws

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$$

$$\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$

Distributivity:

- $A \vee (B \wedge C) \Leftrightarrow (A \vee B) \wedge (A \vee C)$
- $A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$

$$(A \wedge B) \supset C \Leftrightarrow A \supset (B \supset C)$$

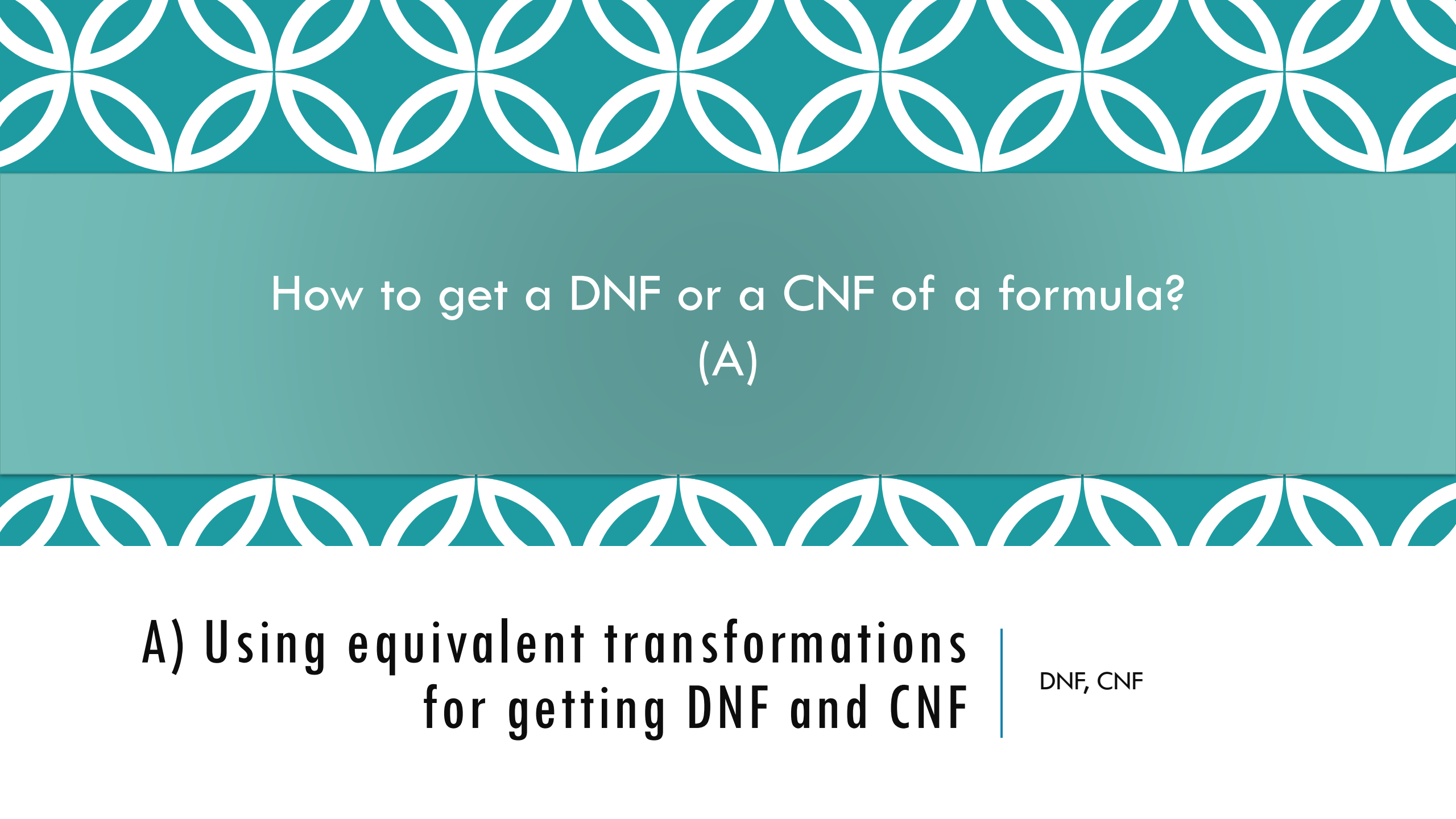
$$A \supset B \Leftrightarrow \neg B \supset \neg A$$

$$A \supset (B \supset C) \Leftrightarrow (A \supset B) \supset (A \supset C)$$

$$(A \vee B) \supset C \Leftrightarrow (A \supset C) \wedge (B \supset C)$$

Expressibility

- $A \supset B \Leftrightarrow \neg(A \wedge \neg B)$
- $A \supset B \Leftrightarrow (\neg A \vee B)$
- $A \wedge B \Leftrightarrow \neg(A \supset \neg B)$
- $A \vee B \Leftrightarrow (\neg A \supset B)$
- $A \vee B \Leftrightarrow \neg(\neg A \wedge \neg B)$
- $A \wedge B \Leftrightarrow \neg(\neg A \vee \neg B)$
- $A \equiv B \Leftrightarrow (A \supset B) \wedge (B \supset A)$



How to get a DNF or a CNF of a formula?
(A)

**A) Using equivalent transformations
for getting DNF and CNF**

DNF, CNF

Find a CNF for the following formula:

$$\varphi = (a \wedge \neg c) \vee \neg(\neg c \rightarrow b).$$

Solution:

$$(a \wedge \neg c) \vee \neg(\neg c \rightarrow b) \Rightarrow (a \wedge \neg c) \vee \neg(\neg \neg c \vee b)$$

$$(a \wedge \neg c) \vee \neg(\neg \neg c \vee b) \Rightarrow (a \wedge \neg c) \vee (\neg \neg \neg c \wedge \neg b)$$

$$(a \wedge \neg c) \vee (\neg \neg \neg c \wedge \neg b) \Rightarrow (a \wedge \neg c) \vee (\neg c \wedge \neg b)$$

$$\begin{aligned} & \overset{A}{(a \wedge \neg c)} \vee \overset{B}{(\neg c \wedge \neg b)} \\ \Rightarrow & \overset{A}{(a \vee (\neg c \wedge \neg b))} \wedge \overset{B}{(\neg c \vee (\neg c \wedge \neg b))} \\ \Rightarrow & ((a \vee \neg c) \wedge (a \vee \neg b)) \wedge ((\neg c \vee \neg c) \wedge (\neg c \vee \neg b)) \\ \Rightarrow & (a \vee \neg c) \wedge (a \vee \neg b) \wedge (\neg c \vee \neg c) \wedge (\neg c \vee \neg b) \end{aligned}$$

$$\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$$

$$(A \supset B) \Leftrightarrow \neg(A \wedge \neg B)$$

$$(A \supset B) \Leftrightarrow (\neg A \vee B)$$

$$(A \wedge B) \Leftrightarrow \neg(A \supset \neg B)$$

$$(A \vee B) \Leftrightarrow (\neg A \supset B)$$

$$(A \vee B) \Leftrightarrow \neg(\neg A \wedge \neg B)$$

$$(A \wedge B) \Leftrightarrow \neg(\neg A \vee \neg B)$$

$$(A \equiv B) \Leftrightarrow ((A \supset B) \wedge (B \supset A))$$

$$(A \vee (B \wedge C)) \Leftrightarrow ((A \vee B) \wedge (A \vee C))$$

$$(A \wedge (B \vee C)) \Leftrightarrow ((A \wedge B) \vee (A \wedge C))$$

$$(A \wedge (B \vee A)) \Leftrightarrow A$$

$$(A \vee (B \wedge A)) \Leftrightarrow A$$

. Simplify (to DNF) the formula $\neg p \wedge \neg q \wedge r \vee \neg p \wedge q \wedge \neg r \vee p \wedge \neg q \wedge \neg r \vee p \wedge \neg q \wedge r$!

- . Simplify (to DNF) the formula $\neg p \wedge \neg q \wedge r \vee \neg p \wedge q \wedge \neg r \vee p \wedge \neg q \wedge \neg r \vee p \wedge \neg q \wedge r$!

Solution: By choosing the first and last elementary conjunction, we get by expansion rule $\neg q \wedge r$; by choosing the third and fourth elementary conjunction we get $p \wedge \neg q$. By choosing $\neg q \wedge r$ and using the absorption rule we can omit the first and last elementary conjunction of the original formula; and by taking $p \wedge \neg q$ we can omit the third and fourth elementary conjunction. So by adding to $\neg q \wedge r$ and $p \wedge \neg q$ the remaining second elementary conjunction we get the result:

$$(\neg q \wedge r) \vee (p \wedge \neg q) \vee (\neg p \wedge q \wedge \neg r)$$

Give the disjunctive normal form of the formula $(p \supset q \supset r) \supset ((p \supset q) \supset r)$!

Give the disjunctive normal form of the formula $(p \supset q \supset r) \supset ((p \supset q) \supset r)$!

Solution: For the sake of easier understanding we make only one simplification in each steps. Who is experienced, can make multiple transformation in parallel in one step. $p \supset (q \supset r) \Leftrightarrow \neg p \vee (\neg q \vee r) \Leftrightarrow \neg p \vee \neg q \vee r$. $((p \supset q) \supset r) \Leftrightarrow \neg(\neg p \vee q) \vee r \Leftrightarrow (p \wedge \neg q) \vee r$. Now we can work with the whole formula: $(\neg p \vee \neg q \vee r) \supset ((p \wedge \neg q) \vee r) \Leftrightarrow \neg(\neg p \vee \neg q \vee r) \vee ((p \wedge \neg q) \vee r) \Leftrightarrow (p \wedge q \wedge \neg r) \vee (p \wedge \neg q) \vee r$.

By applying the distributive rule for formula $(A \wedge x) \vee \neg x$ we get the formula $A \vee \neg x$. Let x be the $\neg r$, $A \wedge x$ the content of the first bracket and apply this rule: $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q) \vee r \Leftrightarrow (p \wedge q) \vee (p \wedge \neg q) \vee r$. By applying the distributive rule in reverse direction for the first two elementary conjunctions we get p , so the original formula logically equivalent with $p \vee r$.

Give the conjunctive normal form of the formula $(p \supset q \supset r) \supset ((p \supset q) \supset r)$!

Give the conjunctive normal form of the formula $(p \supset q \supset r) \supset ((p \supset q) \supset r)$!

Solution: The solution of the previous formula is CNF form, too; hence the solution is $p \vee r$

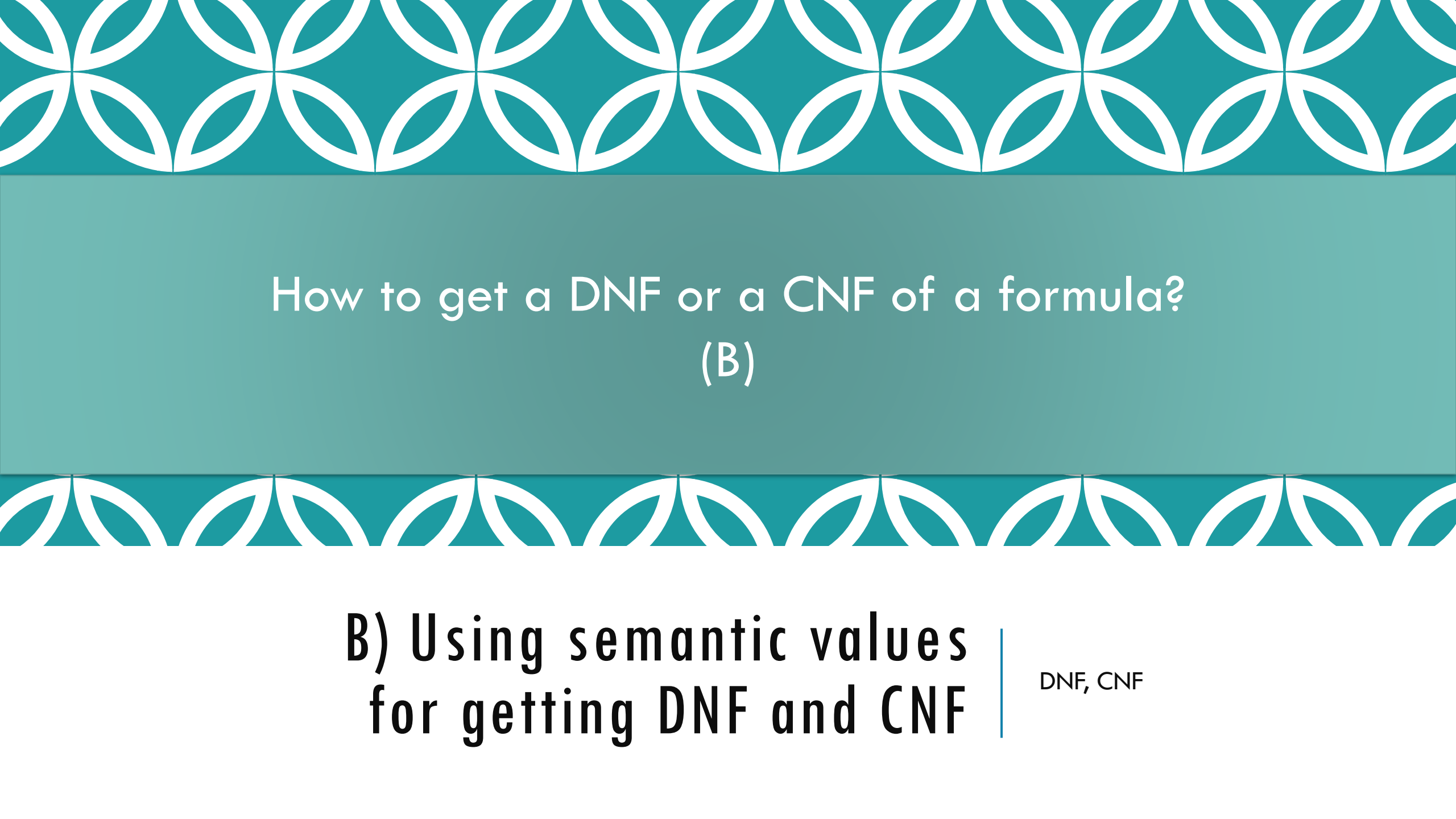
Give the disjunctive normal form of the formula $(p \supset q \supset r) \equiv ((p \supset q) \supset r)!$

Give the disjunctive normal form of the formula $(p \supset q \supset r) \equiv ((p \supset q) \supset r)$!

Solution: $(p \supset q \supset r) \equiv ((p \supset q) \supset r) \Leftrightarrow ((p \supset q \supset r) \supset ((p \supset q) \supset r)) \wedge (((p \supset q) \supset r) \supset (p \supset q \supset r))$ A $((p \supset q \supset r) \supset ((p \supset q) \supset r)) \Leftrightarrow p \vee r$ by the previous exercise, so we work now with the second part of the formula. $((p \supset q) \supset r) \supset (p \supset q \supset r) \Leftrightarrow \neg((p \supset q) \supset r) \vee (p \supset (q \supset r)) \Leftrightarrow \neg((\neg p \vee q) \supset r) \vee (p \supset (\neg q \vee r)) \Leftrightarrow \neg(\neg(\neg p \vee q) \vee r) \vee (\neg p \vee (\neg q \vee r)).$

Let us move the negation to the literals by using the de Morgan rule! $\neg(\neg(\neg p \vee q) \vee r) \vee (\neg p \vee (\neg q \vee r)) \Leftrightarrow ((\neg p \vee q) \wedge \neg r) \vee \neg p \vee \neg q \vee r.$

Let use the trick we used two exercises before! $((\neg p \vee q) \wedge \neg r) \vee \neg p \vee \neg q \vee r \Leftrightarrow (\neg p \vee q) \vee \neg p \vee \neg q \vee r \Leftrightarrow \top.$ By combining this with the previous result we get: $(p \vee r) \wedge \top \Leftrightarrow (p \vee r).$



How to get a DNF or a CNF of a formula?
(B)

**B) Using semantic values
for getting DNF and CNF**

DNF, CNF

Expression formulas in DNF

Should be expressed: $(\neg a \supset b) \wedge (b \wedge c)$

a	b	c	$(\neg a \supset b) \wedge (b \wedge c)$	
0	0	0	0	—
0	0	1	0	—
0	1	0	0	—
0	1	1	1	$\neg a \wedge b \wedge c$
1	0	0	0	—
1	0	1	0	—
1	1	0	0	—
1	1	1	1	$a \wedge b \wedge c$

Result: $(\neg a \wedge b \wedge c) \vee (a \wedge b \wedge c)$

$$(a \wedge \neg c) \wedge ((\neg c \supset b) \vee a)$$

Or in this way (Method 2)

- i. Take the formula Φ (Step 1)
- ii. Construct the DNF of $\neg\Phi$ (Steps 2-4)
- iii. Construct the $\neg(\neg\Phi)$ that is $= \Phi$ (Step 5)
- iv. Apply the De Morgan rules for the $\Phi = \neg(\neg\Phi)$ (Steps 6-7)

$$1 \quad \varphi : \neg(a \wedge \neg c) \vee \neg((\neg c \supset b) \vee a)$$

$$2 \quad \neg\varphi : (a \wedge \neg c) \wedge ((\neg c \supset b) \vee a)$$

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$$

$$\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$

(a)	(b)	(c)	($\neg\varphi$)	
0	0	0	0	—
0	0	1	0	—
0	1	0	0	—
0	1	1	0	—
1	0	0	1	$(a \wedge \neg b \wedge \neg c)$
1	0	1	0	—
1	1	0	1	$(a \wedge b \wedge \neg c)$
1	1	1	0	—

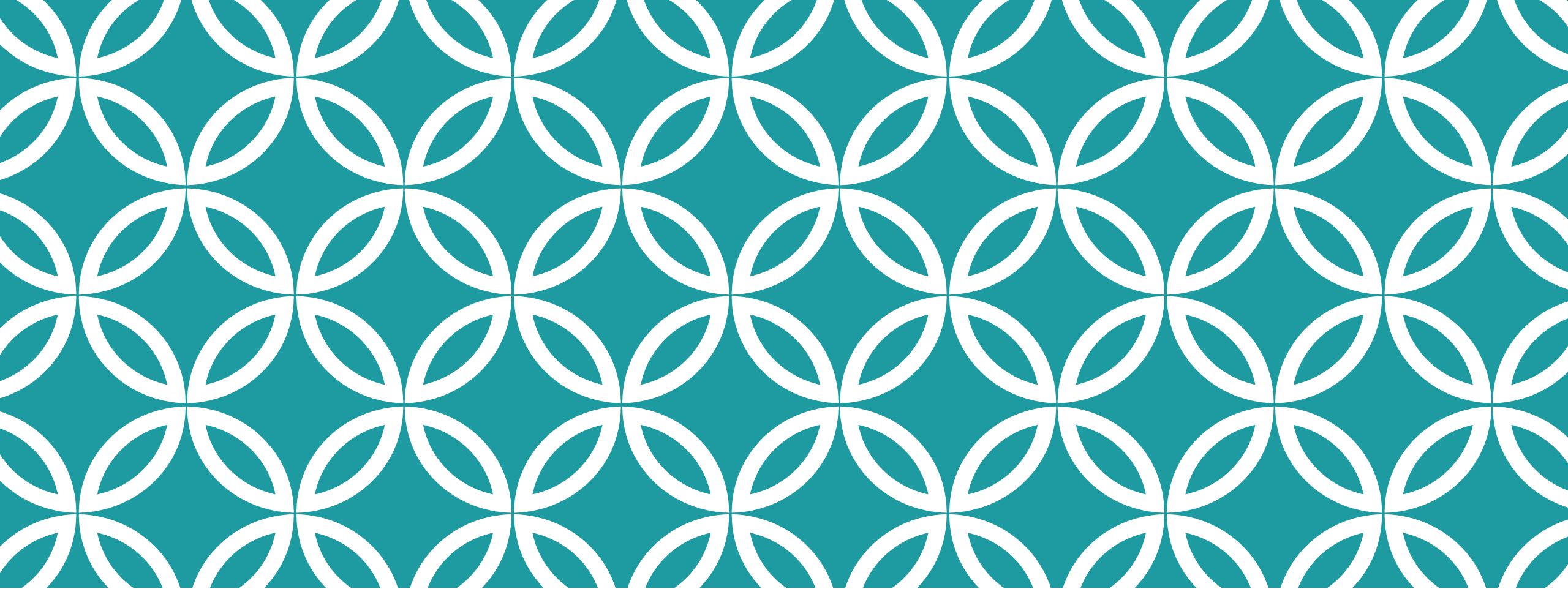
$$4 \quad \neg\varphi \equiv (a \wedge \neg b \wedge \neg c) \vee (a \wedge b \wedge \neg c)$$

$$5 \quad \neg\neg\varphi$$

$$\neg((a \wedge \neg b \wedge \neg c) \vee (a \wedge b \wedge \neg c))$$

$$6 \quad \neg(a \wedge \neg b \wedge \neg c) \wedge \neg(a \wedge b \wedge \neg c)$$

$$7 \quad (\neg a \vee \neg\neg b \vee \neg\neg c) \wedge (\neg a \vee \neg b \vee \neg\neg c)$$



Exercises

DNF, CNF

Exercise

8.1.9. The formula $X \wedge Y$ is the CNF of one or more of the following formulas.
Which is/are this/these formula(s)?

(a) $X \wedge Y \wedge \neg Z$

(b) $\neg(\neg X \vee \neg Y)$

(c) $X \supset (Y \vee \neg X)$

(d) $(X \supset Y) \supset (\neg Y \supset \neg X)$

Solutions:

a) no

b) $\neg\neg X \wedge \neg\neg Y \Leftrightarrow X \wedge Y$

c) $\neg X \vee (Y \vee \neg X) \Leftrightarrow \neg X \vee Y$

d) $(\neg X \vee Y) \supset (\neg\neg Y \vee \neg X) \Leftrightarrow$
 $\neg(\neg X \vee Y) \vee (Y \vee \neg X) \Leftrightarrow$
 $(X \wedge \neg Y) \vee Y \vee \neg X \Leftrightarrow$
 $(X \vee Y) \wedge (\neg Y \vee Y) \vee \neg X \rightarrow$
 $(X \vee Y) \vee \neg X$

that is true for every X and Y :

$$T \vee Y$$

De Morgan Laws

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$$

$$\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$

Distributivity:

- $A \vee (B \wedge C) \Leftrightarrow (A \vee B) \wedge (A \vee C)$
- $A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$

$$(A \wedge B) \supset C \Leftrightarrow A \supset (B \supset C)$$

$$A \supset B \Leftrightarrow \neg B \supset \neg A$$

$$A \supset (B \supset C) \Leftrightarrow (A \supset B) \supset (A \supset C)$$

$$(A \vee B) \supset C \Leftrightarrow (A \supset C) \wedge (B \supset C)$$

Expressibility

- $A \supset B \Leftrightarrow \neg(A \wedge \neg B)$
- $A \supset B \Leftrightarrow (\neg A \vee B)$
- $A \wedge B \Leftrightarrow \neg(A \supset \neg B)$
- $A \vee B \Leftrightarrow (\neg A \supset B)$
- $A \vee B \Leftrightarrow \neg(\neg A \wedge \neg B)$
- $A \wedge B \Leftrightarrow \neg(\neg A \vee \neg B)$
- $A \equiv B \Leftrightarrow (A \supset B) \wedge (B \supset A)$

Exercises: DNF, CNF

- 44. Give a disjunctive normal form of the formulae $(p \supset q) \supset p$!
- 45. Give the conjunctive normal form of the formula $(p \supset q) \supset p$!
- 46. Give the disjunctive normal form of the formula $(p \supset q \supset r) \supset ((p \supset q) \supset r)$!
- 47. Give the conjunctive normal form of the formula $(p \supset q \supset r) \supset ((p \supset q) \supset r)$!

De Morgan Laws

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$$

$$\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$

Distributivity:

- $A \vee (B \wedge C) \Leftrightarrow (A \vee B) \wedge (A \vee C)$
- $A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$

$$(A \wedge B) \supset C \Leftrightarrow A \supset (B \supset C)$$

$$A \supset B \Leftrightarrow \neg B \supset \neg A$$

$$A \supset (B \supset C) \Leftrightarrow (A \supset B) \supset (A \supset C)$$

$$(A \vee B) \supset C \Leftrightarrow (A \supset C) \wedge (B \supset C)$$

Expressibility

- $A \supset B \Leftrightarrow \neg(A \wedge \neg B)$
- $A \supset B \Leftrightarrow (\neg A \vee B)$
- $A \wedge B \Leftrightarrow \neg(A \supset \neg B)$
- $A \vee B \Leftrightarrow (\neg A \supset B)$
- $A \vee B \Leftrightarrow \neg(\neg A \wedge \neg B)$
- $A \wedge B \Leftrightarrow \neg(\neg A \vee \neg B)$
- $A \equiv B \Leftrightarrow (A \supset B) \wedge (B \supset A)$

Exercises: DNF, CNF

48. Give the disjunctive normal form of the formula $(p \supset q \supset r) \equiv ((p \supset q) \supset r)!$
50. Give the disjunctive normal form of the formula $p \wedge q \wedge r \vee (r \wedge (\neg q \vee \neg p)) \vee \neg p \wedge \neg q!$
51. Give the conjunctive normal form of the formula $p \wedge q \wedge r \vee (r \wedge (\neg q \vee \neg p)) \vee \neg p \wedge \neg q!$
52. Give the disjunctive normal form of the formula $\neg p \wedge \neg q \wedge r \vee \neg r \wedge (p \wedge \neg q \vee \neg p \wedge q)!$
53. Give the conjunctive normal form of the formula $\neg p \wedge \neg q \wedge r \vee \neg r \wedge (p \wedge \neg q \vee \neg p \wedge q)!$
54. Give the simplest DNF of the formula $\neg p \wedge q \wedge r \vee p \wedge q \vee q \wedge r!$
55. Give the simplest DNF of the formula $\neg p \wedge q \wedge r \vee p \wedge q \vee q \wedge r!$
56. Give the DNF of the formula $\neg p \wedge \neg q \wedge \neg r \wedge s \vee p \wedge \neg q \wedge \neg r \wedge \neg s \vee r \wedge s \vee \neg p \wedge \neg q \wedge r \wedge \neg s!$
57. Give the CNF of the formula $\neg p \wedge \neg q \wedge \neg r \wedge s \vee p \wedge \neg q \wedge \neg r \wedge \neg s \vee r \wedge s \vee \neg p \wedge \neg q \wedge r \wedge \neg s!$
58. Give the (simplest) DNF of the formula $(p \vee q \vee (r \wedge \neg s) \vee s)!$
59. Give the CNF of the formula $(p \vee q \vee (r \wedge \neg s) \vee s)!$

De Morgan Laws

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$$

$$\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$

Distributivity:

- $A \vee (B \wedge C) \Leftrightarrow (A \vee B) \wedge (A \vee C)$
- $A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$

$$(A \wedge B) \supset C \Leftrightarrow A \supset (B \supset C)$$

$$A \supset B \Leftrightarrow \neg B \supset \neg A$$

$$A \supset (B \supset C) \Leftrightarrow (A \supset B) \supset (A \supset C)$$

$$(A \vee B) \supset C \Leftrightarrow (A \supset C) \wedge (B \supset C)$$

Expressibility

- $A \supset B \Leftrightarrow \neg(A \wedge \neg B)$
- $A \supset B \Leftrightarrow (\neg A \vee B)$
- $A \wedge B \Leftrightarrow \neg(A \supset \neg B)$
- $A \vee B \Leftrightarrow (\neg A \supset B)$
- $A \vee B \Leftrightarrow \neg(\neg A \wedge \neg B)$
- $A \wedge B \Leftrightarrow \neg(\neg A \vee \neg B)$
- $A \equiv B \Leftrightarrow (A \supset B) \wedge (B \supset A)$

Disjunctive normal form

31. Give the full disjunctive normal form for which the truth table is the following: $f(p, q) = (0010)!$

$$p \wedge \neg q$$

Exercises

30. Give the full disjunctive normal form for which the truth table is the following:
 $f(p, q) = (1011)!$
31. Give the full disjunctive normal form for which the truth table is the following:
 $f(p, q) = (0010)!$
32. Give the full disjunctive normal form for which the truth table is the following:
 $f(p, q, r) = (10011000)!$

$$\neg(A \vee B) \Leftrightarrow (\neg A \wedge \neg B)$$

$$\neg(A \wedge B) \Leftrightarrow (\neg A \vee \neg B)$$

$$(A \supset B) \Leftrightarrow \neg(A \wedge \neg B)$$

$$(A \supset B) \Leftrightarrow (\neg A \vee B)$$

$$(A \wedge B) \Leftrightarrow \neg(A \supset \neg B)$$

$$(A \vee B) \Leftrightarrow (\neg A \supset B)$$

$$(A \vee B) \Leftrightarrow \neg(\neg A \wedge \neg B)$$

$$(A \wedge B) \Leftrightarrow \neg(\neg A \vee \neg B)$$

$$(A \equiv B) \Leftrightarrow ((A \supset B) \wedge (B \supset A))$$

$$(A \vee (B \wedge C)) \Leftrightarrow ((A \vee B) \wedge (A \vee C))$$

$$(A \wedge (B \vee C)) \Leftrightarrow ((A \wedge B) \vee (A \wedge C))$$

$$(A \wedge (B \vee A)) \Leftrightarrow A$$

$$(A \vee (B \wedge A)) \Leftrightarrow A$$